E-COMMERCE APPLICATION DEVELOPMENT INTERNSHIP

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COMPANY NAME: COGNIZANT TECHNOLOGY SOLUTIONS

INTERNSHIP DURATION: 31/01/2026 – 04/05/2026

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INTRODUCTION

- This internship is part of **GenC training**, which prepares fresh graduates for real-world IT roles.
- Domain: **E-commerce application development**.
- Purpose:
  - To understand how online retail platforms are designed, developed, and deployed.
  - To gain practical exposure to **microservices architecture, cloud deployment, and containerization**.
  - To strengthen both backend (Java Spring Boot) and frontend (React, HTML, CSS, JS, jQuery, Bootstrap) skills.
  - To apply theoretical knowledge from academics into industry-standard projects.

ABOUT THE COMPANY

- **Overview:** Cognizant is a Fortune 500 multinational IT services and consulting company.
- **Location:** Headquarters in Teaneck, New Jersey, USA; major delivery centers in India, including **Siruseri, Chennai(current internship location)**.
- **Services/Products:** Digital transformation, IT consulting, software development, cloud services, AI, and e-commerce solutions.
- **Industry Domain:** Information Technology, Consulting, and Business Process Services.
- **Reputation:** Known for innovation, client partnerships, and large-scale enterprise solutions.

ORGANIZATIONAL STRUCTURE

- Internship under **Programmer Analyst Trainee (GenC)** role.
- Reporting to mentor/supervisor in the **E-commerce solutions team**.
- Exposure to:
- **Agile methodology:** sprint planning, daily stand-ups, retrospectives.
- Collaboration with developers, testers, and project managers.
- Structured workflow: requirement gathering → design → development → testing → deployment.
- Learned how teams are organized into **frontend, backend, DevOps, and QA units**.

INTERNSHIP OBJECTIVES

- Learn practical e-commerce application development.
- Understand **microservices architecture** and service communication.
- Strengthen programming skills in **Java Spring Boot** (backend) and **React** (frontend).
- Apply theoretical knowledge to real-world projects.
- Gain exposure to **DevOps practices** and containerization with Docker.
- Learn cloud deployment strategies (AWS/Azure).
- Improve teamwork, communication, and professional presentation skills.

TOOLS AND TECHNOLOGIES USED

- **Programming Languages:** Java, JavaScript
- **Frameworks:** Spring Boot (backend), React (frontend)
- **Web Technologies:** HTML, CSS, jQuery, Bootstrap
- **Architecture:** Microservices (service discovery, API gateway)
- **Containerization/Cloud:** Docker, Cloud stack (AWS/Azure)
- **Database:** MySQL / Oracle
- **Tools:** GitHub (version control), VS Code (IDE), JIRA (project management), Postman (API testing), JUnit (unit testing)

TASKS ASSIGNED / WORK DONE SO FAR

- **Week 1 (Jan 31 – Feb 7):**
  - Company orientation and introduction to **GenC training modules**.
  - Overview of Cognizant Siruseri campus, policies, and training structure.
  - Initial sessions on software development lifecycle and agile methodology.
- **Week 2 (Feb 8 – Feb 14):**
  - Setup of development environment (Spring Boot, Git, VS Code).
  - Hands-on practice with **Git version control** (clone, commit, push, branch).
  - Introduction to **HTML, CSS, JavaScript** for frontend basics.
- **Week 3 (Feb 15 – Feb 21):**
  - Learning **jQuery** for DOM manipulation and event handling.
  - Introduction to **React**: components, props, and state management.
  - Practice building small UI components with React.

WORK DONE SO FAR

- **Week 4 (Feb 22 – Feb 28):**
- Backend training with **Java Spring Boot**.
- Implemented **Hibernate CRUD operations** (Create, Read, Update, Delete).
- Connected Spring Boot application to a relational database (MySQL).
- **Week 5 (Mar 1 – Mar 7):**
- Consolidation of frontend and backend concepts.
- Mini-assignments to integrate React frontend with Spring Boot backend APIs.
- Debugging exercises and peer code reviews.
- **Week 6 (Mar 8 – Mar 13):**
- **Project assigned in the current week** (E-commerce track).
- Initial requirement gathering and understanding project scope.
- Planning modules and identifying technologies to be used (Spring Boot, React, Docker, Microservices).

LEARNING OUTCOMES

- Learned **Git version control**: cloning repositories, committing changes, pushing to remote, branching.
- Gained confidence in **HTML and CSS** for structuring and styling web pages.
- Practiced **JavaScript fundamentals**: functions, events, DOM manipulation.
- Understood **jQuery** for simplifying DOM handling and AJAX requests.
- Implemented **CRUD operations in Spring Boot with Hibernate** (Create, Read, Update, Delete).
- Connected backend application to a relational database (MySQL).
- Improved debugging skills and ability to troubleshoot environment setup issues.
- Learned to work in a structured training environment under **GenC program**.

CHALLENGES FACED

- Difficulty in setting up **Spring Boot environment** and configuring Hibernate.
- Understanding database mappings and relationships in Hibernate.
- Initial confusion with Git branching and resolving merge conflicts.
- Adapting to new tools like JIRA and Postman for project tracking and API testing.
- Managing time between multiple modules (frontend basics + backend CRUD).

HOW THEY WERE OVERCOME

- Took guidance from mentor and peers during training sessions.
- Referred to **official documentation** and Cognizant learning portals.
- Practiced small coding exercises to strengthen CRUD concepts.
- Used debugging tools and console logs to resolve backend issues.
- Learned Git branching strategies and practiced collaborative workflows.

CONCLUSION

- Internship under **GenC training at Cognizant Siruseri** has provided a strong foundation in **frontend basics and backend development**.
- Progressed steadily from orientation → Git → HTML/CSS/JS → jQuery → Spring Boot + Hibernate CRUD.
- Project has been assigned in the current week, marking the transition from training modules to **real-world application development**.
- Review–1 demonstrates readiness to move from learning modules into project execution.
- Internship is helping bridge the gap between academic knowledge and industry practices, while improving both technical and professional skills.